在不使用 Upgrade Advisor 进行升级之前，应验证哪些内容？

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在不使用 Upgrade Advisor 进行升级之前，应验证哪些内容？

升级前要验证的内容

如果不使用 "Active IQ" Upgrade Advisor 要规划升级，您应在升级之前验证集群升级限制和集群活动。

验证集群升级限制

如果不使用 "Active IQ" Upgrade Advisor ，则需要验证集群是否未超过平台系统限制。除了平台系统限制之外， SAN 还具有一些限制，您还应进行验证。

1. 验证集群是否未超过您平台的系统限制。

   "NetApp Hardware Universe"

2. 如果您的集群配置了 SAN ，请验证它是否未超过 FC ， FCoE 和 iSCSI 的配置限制。

   "NetApp Hardware Universe"

3. 确定 CPU 和磁盘利用率： `node run -node node_name -command sysstat -c 10 -x 3`

   您应监控 CPU 和磁盘利用率 30 秒。对于报告的所有 10 个测量值， "CPU* " 和 "* 磁盘利用率 * " 列中的值不应超过 50% 。在升级完成之前，不应向集群添加额外负载。注意： CPU 和磁盘利用率可能会在您的环境中的不同时间有所不同。因此，最好在预期升级窗口的时间段内检查 CPU 和磁盘利用率。

验证当前集群活动

如果不使用 "Active IQ" Upgrade Advisor ，在升级之前，您应手动验证是否未运行任何作业，以及是否终止任何不持续可用的 CIFS 会话。

确认没有正在运行的作业

在升级 ONTAP 软件之前，您必须验证集群作业的状态。如果存在任何聚合，卷， NDMP （转储或还原）或 Snapshot 作业（例如创建，删除，移动，修改，复制， 和挂载作业）正在运行或排队，您必须允许作业成功完成或停止已排队的条目。

1. 查看任何正在运行或已排队的聚合，卷或 Snapshot 作业的列表： `job show`

```
cluster1::> job show
                            Owning
Job ID Name                 Vserver    Node           State
------ -------------------- ---------- -------------- ----------
8629   Vol Reaper           cluster1   -              Queued
       Description: Vol Reaper Job
8630   Certificate Expiry Check
                            cluster1   -              Queued
       Description: Certificate Expiry Check
.
.
.
```

2. 如果有任何正在运行的作业，请允许它们成功完成。
3. 删除任何已排队的聚合，卷或 Snapshot 副本作业： `job delete -id job_id`

```
cluster1::> job delete -id 8629
```

4. 确认没有聚合，卷或 Snapshot 作业正在运行或已排队： `job show`

   在此示例中，已删除所有正在运行和已排队的作业：

```
cluster1::> job show
                            Owning
Job ID Name                 Vserver    Node           State
------ -------------------- ---------- -------------- ----------
9944   SnapMirrorDaemon_7_2147484678
                            cluster1   node1          Dormant
       Description: Snapmirror Daemon for 7_2147484678
18377  SnapMirror Service Job
                            cluster1   node0          Dormant
       Description: SnapMirror Service Job
2 entries were displayed
```

确定应终止的活动 CIFS 会话

在升级 ONTAP 软件之前，您应确定并正常终止所有不持续可用的 CIFS 会话。

持续可用的 CIFS 共享由 Hyper-V 或 Microsoft SQL Server 客户端使用 SMB 3.0 协议访问，无需在升级之前终止。

1. 确定任何已建立但不持续可用的 CIFS 会话： `vserver cifs session show -continuously -available Yes -instance`

此命令可显示有关任何不具有持续可用性的 CIFS 会话的详细信息。在继续 ONTAP 升级之前，您应先终止它们。

```
cluster1::> vserver cifs session show -continuously-available Yes
-instance

                        Node: node1
                     Vserver: vs1
                  Session ID: 1
                Connection ID: 4160072788
Incoming Data LIF IP Address: 198.51.100.5
      Workstation IP address: 203.0.113.20
    Authentication Mechanism: NTLMv2
                Windows User: CIFSLAB\user1
                   UNIX User: nobody
                 Open Shares: 1
                  Open Files: 2
                  Open Other: 0
              Connected Time: 8m 39s
                   Idle Time: 7m 45s
            Protocol Version: SMB2_1
      Continuously Available: No
1 entry was displayed.
```

2. 如有必要，请确定为您标识的每个 CIFS 会话打开的文件： `vserver cifs session file show -session-id session_ID`

```
cluster1::> vserver cifs session file show -session-id 1

Node:       node1
Vserver:    vs1
Connection: 4160072788
Session:    1
File    File      Open Hosting
Continuously
ID      Type      Mode Volume          Share                 Available
------- --------- ---- --------------- ---------------------
------------
1       Regular   rw   vol10           homedirshare          No
Path: \TestDocument.docx
2       Regular   rw   vol10           homedirshare          No
Path: \file1.txt
2 entries were displayed.
```

如何在 ONTAP 升级期间更新固件

由于升级 ONTAP 包括升级固件，因此您无需手动更新固件。执行 ONTAP 升级时，ONTAP 升级软件包附带的集群固件将复制到每个节点的启动设备，并自动安装新固件。

如果集群中的固件版本早于 ONTAP 升级软件包捆绑的固件，则会自动更新以下组件的固件：

- BIOS/加载程序
- 服务处理器(SP)或基板管理控制器(BMC)
- 存储架
- Disk
- Flash Cache

如果需要，您还可以在 ONTAP 升级之间手动更新固件。

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